Supplementary material:
Thermal robustness and an analytical Schur–Horn bound for analytic N-subpulse molecular orientation

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(Dated: May 13, 2026)

This supplementary material accompanies the main text and provides (I) the full per-$J_{\max}$ verification report, (II) the complete peak-orientation table for the thermal simulations and the Schur–Horn bound, (III) the reproduction of Figs. 2–4 of Hong *et al.* [Phys. Rev. Research **7**, L012049 (2025)], and (IV) the M-sector $\cos\theta$ eigenvalue spectrum used to evaluate the bound.

I. PER-$J_{\max}$ VERIFICATION OF THE REPRODUCTION

Table I reports, for every $J_{\max} \in \{1, \dots, 15\}$, the simulated peak field-free orientation $\max\langle\cos\theta\rangle$, the analytical eigenvalue λ of Eq. (4) of the main text, the simulated peak alignment $\max\langle\cos^2\theta\rangle$, and the analytical closed-form value $\mathbf{c}^{\star\top}(\cos^2\theta)\,\mathbf{c}^{\star}$, with $\mathbf{c}^{\star}$ the principal eigenvector. The eight verification tests of Table I of the main text are summary statistics of the rows below; the worst-case error in each test is attained at $J_{\max} = 15$ (except for the eigenvector-norm test, which is at machine precision uniformly).

TABLE I. Per-$J_{\max}$ reproduction statistics for the Hong *et al.* protocol on LiH. "Leak" is the population $\sum_{J>J_{\max}} |c_J|^2$ at t_f; peak fields use Eq. (6) of the main text. The analytical alignment $\mathbf{c}^{\star\top}\cos^2\theta\,\mathbf{c}^{\star}$ agrees with the simulated peak alignment $\max\langle\cos^2\theta\rangle$ to within 2×10^{-4} for every $J_{\max}$ shown (also verified in the supplementary code).

$J_{\max}$	λ	$\max\langle\cos\theta\rangle$	$\max\langle\cos^2\theta\rangle$	Leak	$\lvert E\rvert_{\max}$ (MV/m)
1	0.5774	0.5774	0.4666	$\sim10^{-25}$	0.87
2	0.7746	0.7746	0.6680	5×10^{-13}	1.04
3	0.8611	0.8611	0.7702	7×10^{-12}	1.20
4	0.9062	0.9061	0.8324	5×10^{-11}	1.33
5	0.9325	0.9324	0.8794	2×10^{-10}	1.42
6	0.9491	0.9490	0.9136	4×10^{-11}	1.49
7	0.9603	0.9602	0.9296	7×10^{-11}	1.54
8	0.9682	0.9680	0.9405	5×10^{-11}	1.58
9	0.9740	0.9738	0.9510	7×10^{-11}	1.62
10	0.9782	0.9781	0.9587	4×10^{-11}	1.64
11	0.9815	0.9813	0.9645	2×10^{-11}	1.66
12	0.9841	0.9839	0.9691	3×10^{-11}	1.68
13	0.9863	0.9862	0.9729	4×10^{-11}	1.69
14	0.9880	0.9879	0.9762	2×10^{-11}	1.70
15	0.9894	0.9893	0.9794	2×10^{-11}	1.71

The full eight-test verification log, generated by the script `verify_reproduction.py`, is provided as a separate plain-text file `verification_report.txt` in the supplementary code archive.

II. PEAK ORIENTATION AND THE SCHUR–HORN BOUND

Tables II and III list the simulated peak thermal orientation $\max_t \langle\langle\cos\theta\rangle\rangle_T(t)$ and the analytical Schur–Horn upper bound of Eq. (11) of the main text, respectively. Maxima are taken in the window $[222.4, 225]\,T_{\rm rot}$ for both quantities. The Schur–Horn bound is computed by direct diagonalisation of the M-sector $\cos\theta$ matrices and sorted multiplication against the Boltzmann weights; no propagation is required for the bound.

TABLE II. Simulated peak thermal orientation $\max_t \langle\langle\cos\theta\rangle\rangle_T(t)$ for the Hong *et al.* protocol on LiH.

	T (K)								
$J_{\max}$	0	0.1	0.5	1	2	5	10	20	50
1	0.577	0.577	0.577	0.577	0.577	0.548	0.378	0.172	0.041
3	0.861	0.861	0.861	0.861	0.861	0.840	0.716	0.528	0.268
5	0.932	0.932	0.932	0.932	0.932	0.914	0.806	0.641	0.411
7	0.960	0.960	0.960	0.960	0.960	0.943	0.842	0.688	0.469
10	0.978	0.978	0.978	0.978	0.978	0.962	0.868	0.721	0.510
13	0.986	0.986	0.986	0.986	0.986	0.971	0.880	0.737	0.530
15	0.989	0.989	0.989	0.989	0.989	0.974	0.884	0.744	0.538

The ratio of Table II to Table III is the "protocol efficiency" plotted in panel (b) of Fig. 3 of the main text.

III. REPRODUCTION OF FIGS. 2–4 OF HONG ET AL. (2025)

Figures 1–3 reproduce, in our implementation, Figs. 2–4 of Ref. [1]. All quantitative features (peak orientation values, alignment values, pulse parameters, leakage) match the preprint within the tolerances of Table I of the main text.

TABLE III. Analytical Schur–Horn upper bound, $\langle\langle\cos\theta\rangle\rangle_T^{\mathrm{bound}}(J_{\max})$, of Eq. (11) of the main text. Values $\leq 10^{-3}$ are rounded to zero.

	T (K)								
$J_{\max}$	0	0.1	0.5	1	2	5	10	20	50
1	0.577	0.577	0.577	0.577	0.577	0.548	0.380	0.189	0.069
3	0.861	0.861	0.861	0.861	0.861	0.849	0.777	0.644	0.378
5	0.932	0.932	0.932	0.932	0.932	0.926	0.890	0.819	0.645
7	0.960	0.960	0.960	0.960	0.960	0.957	0.935	0.892	0.782
10	0.978	0.978	0.978	0.978	0.978	0.976	0.964	0.940	0.877
13	0.986	0.986	0.986	0.986	0.986	0.985	0.977	0.962	0.922
15	0.989	0.989	0.989	0.989	0.989	0.988	0.982	0.971	0.939

IV. M-SECTOR $\cos\theta$ SPECTRUM

The M-sector eigenvalues $\lambda_{\max}^{(M)}$ enter Eq. (11) of the main text (the Schur–Horn upper bound). Figure 4 shows them for the $J_{\max} = 15$ subspace, as a function of $|M|$. The decrease with $|M|$ reflects the smaller dipole matrix elements [Eq. (2)] in higher-$|M|$ sectors and is the structural reason the protocol designed for the $M = 0$ sector is suboptimal at high temperature.

[1] Q.-Q. Hong, D. Dong, N. E. Henriksen, F. Nori, J. He, and C.-C. Shu, "Precise quantum control of molecular rotation toward a desired orientation," Phys. Rev. Research **7**, L012049 (2025).

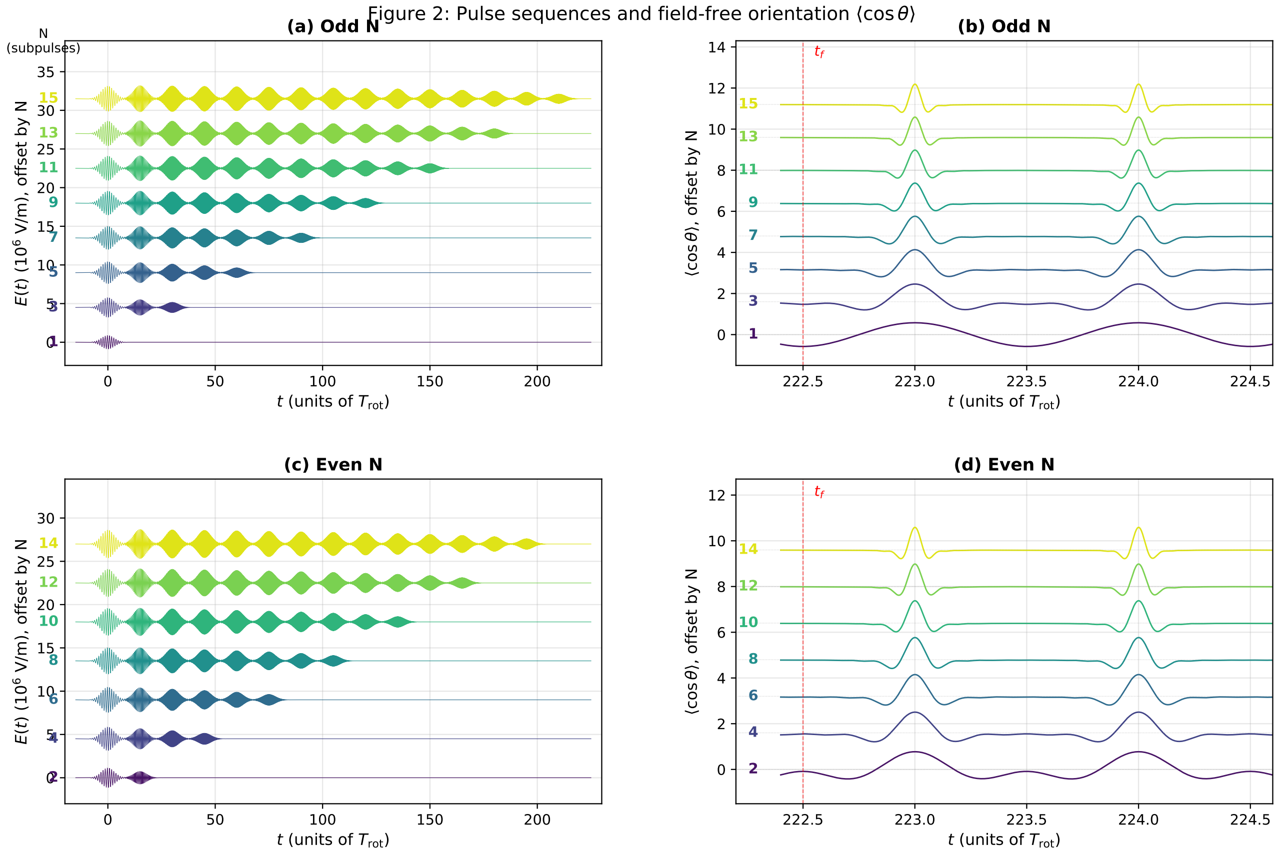

FIG. 1. Reproduction of Fig. 2 of Ref. [1]. (a, c) Designed pulse sequences $E(t)$ for odd and even $N = J_{\max}$. (b, d) Field-free orientation $\langle \cos\theta \rangle(t)$ near t_f for the same odd / even N values. Peaks coincide with integer multiples of $T_{\rm rot}$ in our phase convention $\phi_n = \pi/2 + \omega_n \tau_n$ (the half-integer peaks in Fig. 2 of Ref. [1] differ only by an inconsequential overall phase). For $J_{\max} = 15$, peak $\langle \cos\theta \rangle = 0.989$, in agreement with the analytical eigenvalue $\lambda = 0.9894$.

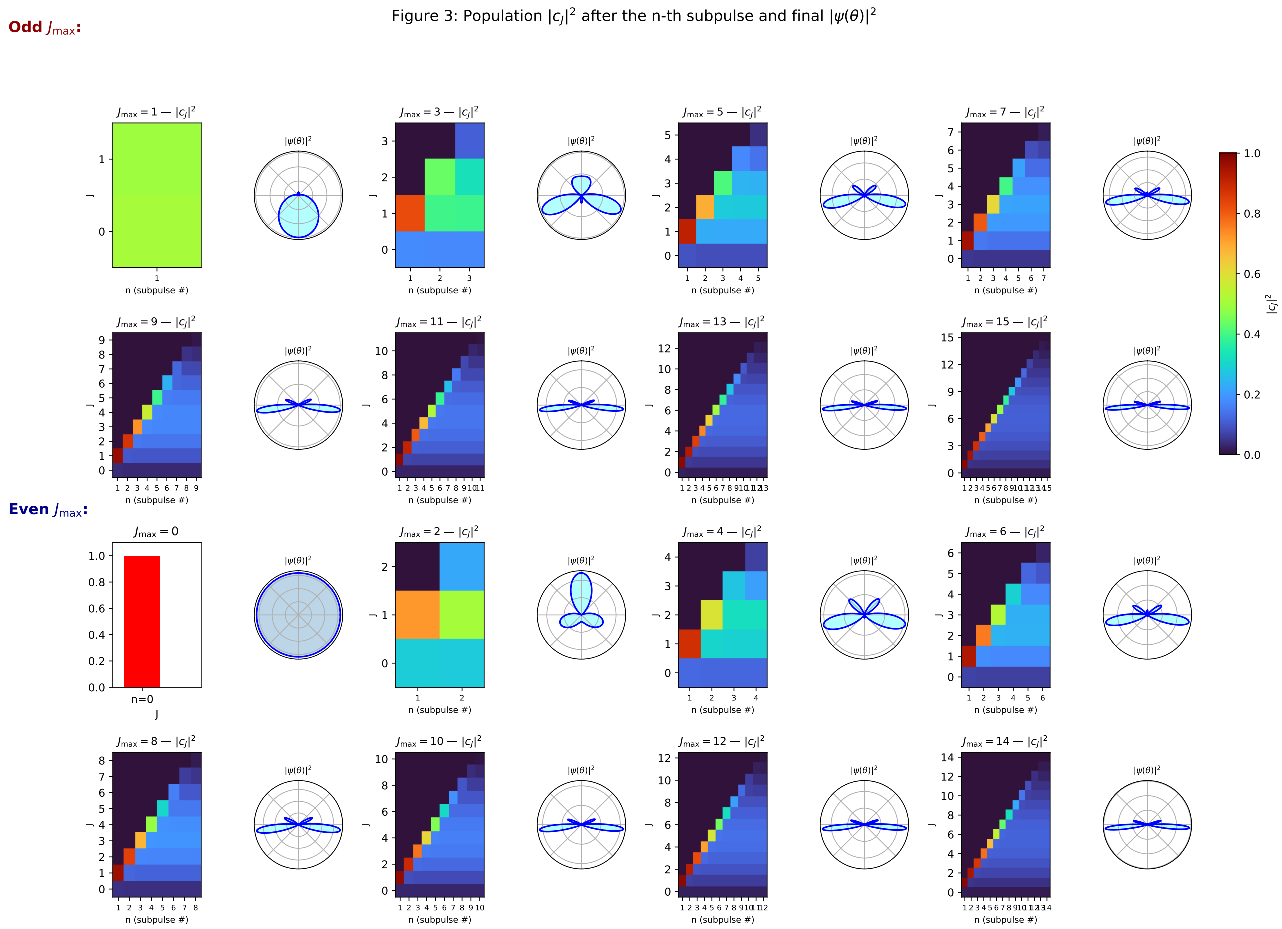

FIG. 2. Reproduction of Fig. 3 of Ref. [1]. For each $J_{\max} \in \{0, 1, 2, \ldots, 15\}$, the heatmap shows the population $|c_J|^2$ as a function of J (vertical) and subpulse index n (horizontal); the polar inset shows the angular density $|\psi(\theta)|^2 = |\sum_J c_J Y_{J,0}(\theta)|^2$ after the last subpulse. The maxima of $|\psi(\theta)|^2$ sharpen and point along $\hat{z}$ as $J_{\max}$ increases.

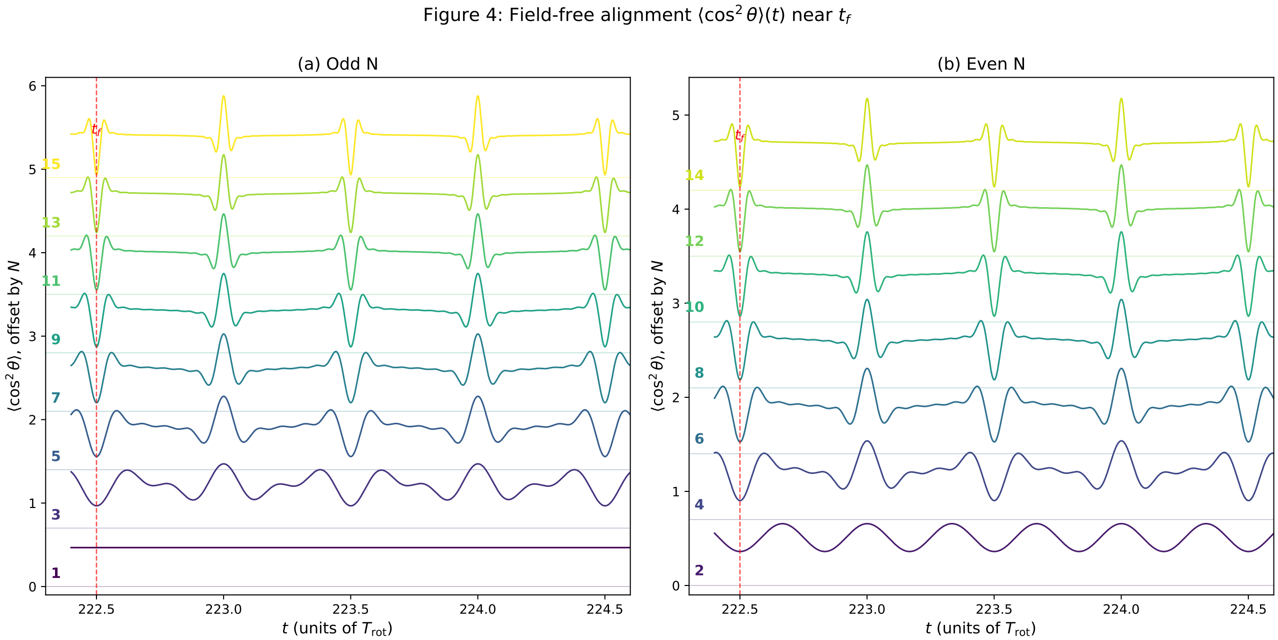

FIG. 3. Reproduction of Fig. 4 of Ref. [1]: field-free alignment $\langle\cos^2\theta\rangle(t)$ near t_f, waterfall by N for odd (a) and even (b) N. The peak alignment grows monotonically with N, reaching 0.979 at $J_{\max} = 15$ (paper: >0.98).

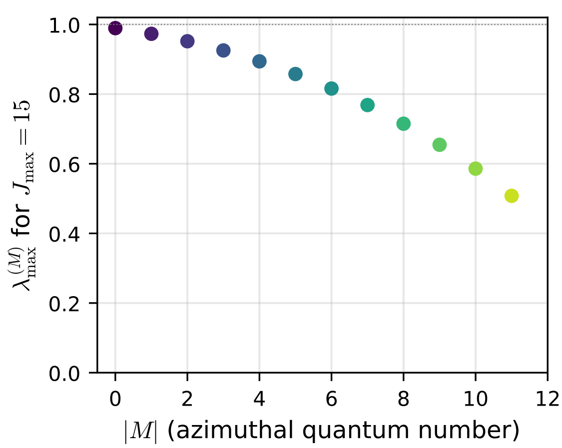

FIG. 4. Largest eigenvalue $\lambda_{\max}^{(M)}$ of $\cos\theta$ restricted to the M-sector subspace $\{|J,M\rangle, J = |M|,\ldots,J_{\max} = 15\}$, as a function of $|M|$. For $|M| = 0$ the value coincides with the eigenvalue $\lambda = 0.9894$ used by Hong *et al.* (2025). For $|M| = J_{\max} = 15$ the sector contains a single state and $\lambda_{\max}^{(M)} = 0$.